HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Produced accurate **engineering drawings** and **Process and Instrumentation Diagrams (P&IDs;)** for water treatment skids, adhering to company procedures and ensuring traceability of supplier documentation.
- Generated equipment, valve, and instrument schedules for multiple projects, collaborating with service departments to define spares and consumables lists for operational handover.
- Assembled comprehensive **Water Treatment System O&M; Documentation and Certification** turnover packs for client delivery and internal service department commissioning engineers.
- Developed and executed **Installation Qualification (IQ) and Operational Qualification (OQ) protocols** to rigorously test water treatment systems against contract requirements and company procedures.
- Managed the collation and transfer of all project documentation to central archives post-project execution, ensuring accessibility and compliance with QA procedures.
- Contributed to the technical review of proposals and designs during the contract stage, verifying compliance with client specifications, mechanical/hydraulic requirements, and procurement budgets.
- Containerized Java Spring Boot microservices using **Docker** and deployed them to an **AWS EC2** environment managed via **Jenkins** CI/CD pipelines, enabling automated testing and deployment cycles.

SKILLS

Engineering Drawing & Design – AutoCAD, Process Flow Schematics, Plant Room Layouts, General Arrangement Drawings

Project Documentation – O&M; Manuals, Certification, Validation Protocols, IQ/OQ, Handover Packs, Archiving

Project Management Support – Scheduling, Procurement Lists, Contract Review, Proposal Support, Technical Estimation, QA Procedures

Software Development – Java, Spring Boot, REST APIs, Hibernate, JPA, SQL, Python, Docker, AWS

Data Analysis & Visualization – Power BI, DAX, Tableau, R, pandas, NumPy

PROJECTS

Water Treatment System Design & Documentation Automation

- Developed automated scripts in **Python** to parse and validate supplier documentation against project requirements, reducing manual data entry time by **40%**.
- Engineered a standardized template system for **IQ/OQ protocols** using advanced Word features and VBA, ensuring consistency and reducing protocol generation time by **30%** across a project portfolio.
- Designed and implemented a streamlined workflow for collating and archiving project documentation using **Jira** and **Confluence**, improving retrieval times for service and compliance teams.

Enterprise Java Application for Environmental Monitoring

- Built a **Java Spring Boot** application to process and monitor environmental sensor data, creating REST APIs for data ingestion and retrieval.
- Implemented data persistence using **Hibernate** and **PostgreSQL**, managing complex relationships between sensor readings, device metadata, and alert configurations.
- Developed comprehensive unit and integration tests using **JUnit** and **Mockito**, achieving over **85%** code coverage for critical application modules.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)